

Beyond the GPU

Driving factors, architectures, and economics of AI hardware acceleration — a systems lecture

In-depth lecture • ~2 hours

Seven movements

1

Why offload

The physical driving factors.

2

Workload pressures

The measured bounds.

3

The framework

Contracts + necessity test.

4

The families

Eight contract changes.

5

Landscape

Grouped, not cataloged.

6

Substrate

Software, packaging, cost.

7

Futures

Graded by evidence.

★

Conclusions

What holds everywhere.

The hardware vocabulary, in software terms



GPU

Thousands of parallel lanes; fast on regular, batched, streamed work.



Matrix / tensor engine

A whole matrix tile per instruction; control cost amortized over many multiply-adds.



Systolic array

A grid where operands flow to neighbors — a value fetched once is reused many times.



HBM & scratchpad

Stacked DRAM for bandwidth; software-managed SRAM for reuse under compiler control.



NPU / dataflow / PIM

Blocks that change a contract: wake-scope energy, static schedule, or compute in memory.



TOPS vs contract

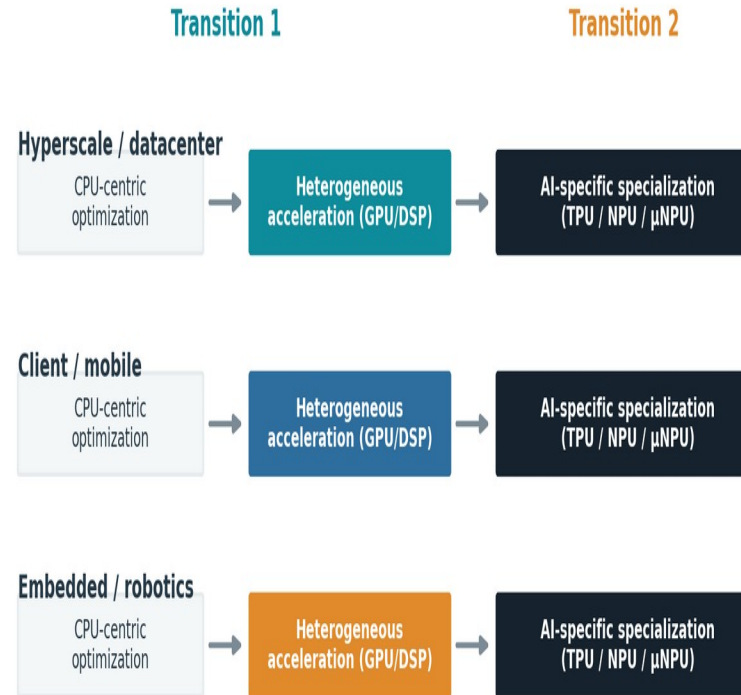
TOPS is a peak rating (marketing). A contract is what hardware promises software.

01

Why work leaves the GPU

The physical pressures behind offload

Two transitions, across the whole continuum



⇒ Software succeeds until it hits a limit; each transition adds specialization rather than replacing what came before.

Why general-purpose scaling stalled



The memory wall

Compute outran memory; the question became which data is resident and reused, not raw latency.



The power wall

Frequency scaling ended; more transistors help only when software exposes parallelism.



Dark silicon

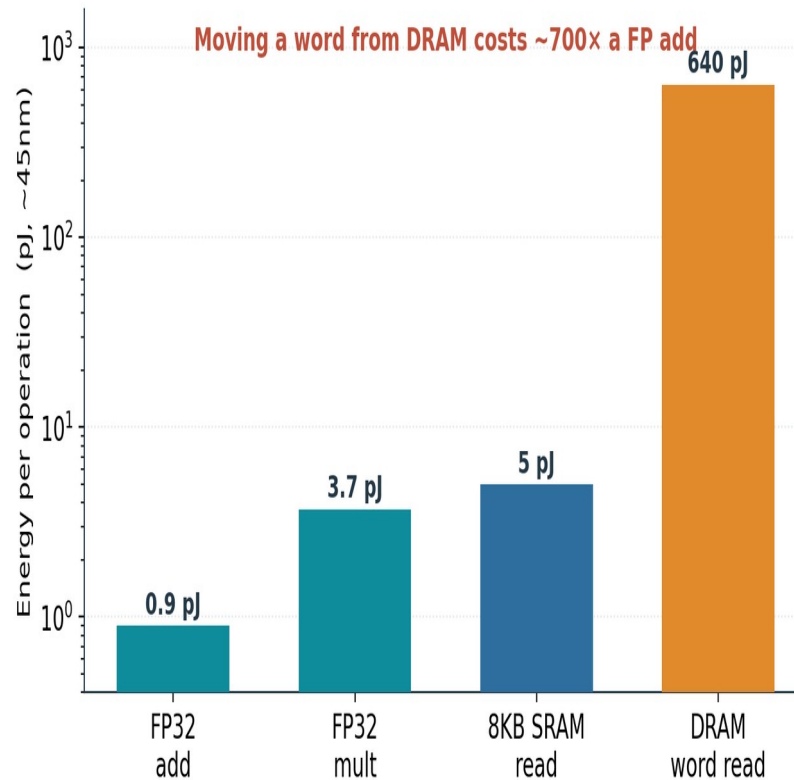
Not all transistors can switch at once in a fixed power budget — specialization does more per joule.



Data movement

Energy is dominated by moving operands, not computing on them — the recurring lever.

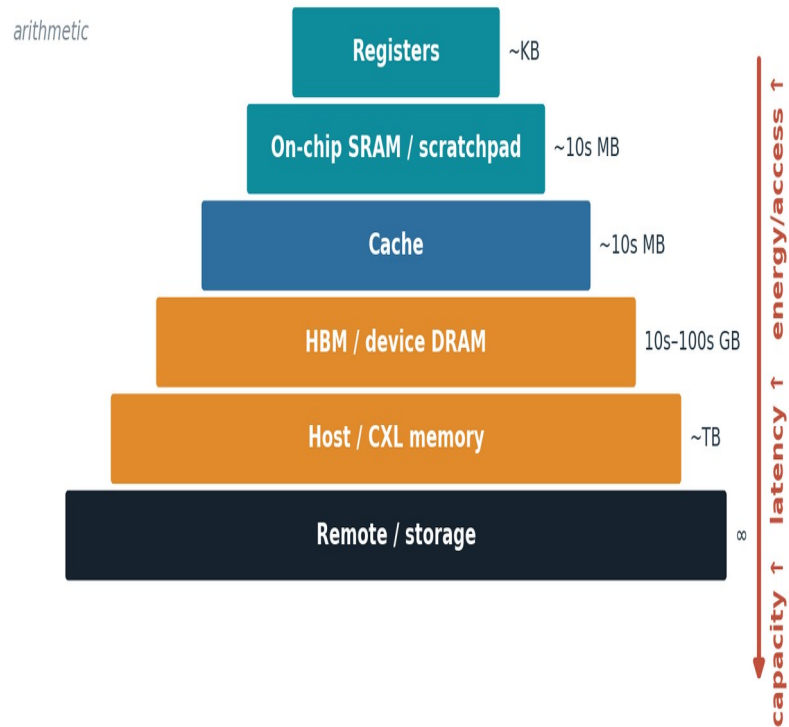
Data movement dominates energy



- Compute is nearly free.
- Moving operands is not.
- Locality sets efficiency, not FLOPS.

⇒ Every family that follows is a different way to move less data, or move it a shorter distance.

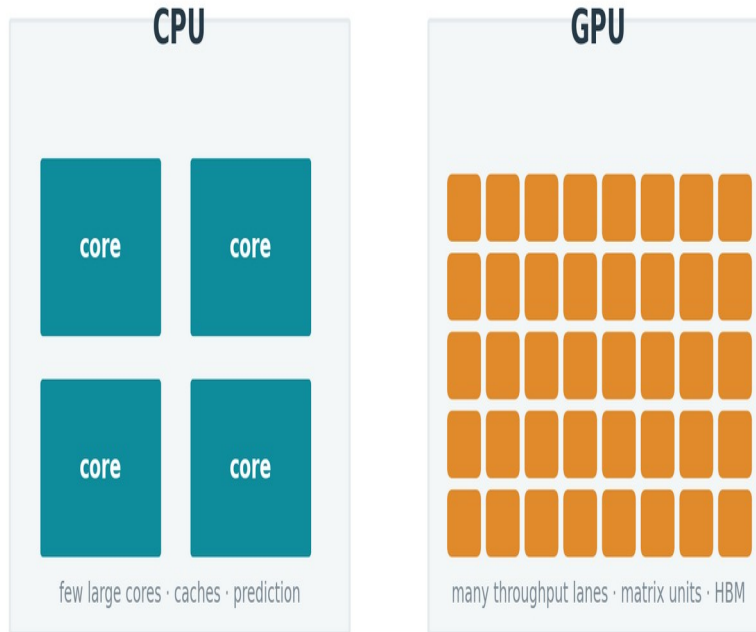
Memory is a distance hierarchy



- Farther = more capacity.
- Also more latency and energy.
- Caches help only with reuse.

⇒ 'Memory-bound' is many distinct problems — each a lever a specialized design can pull.

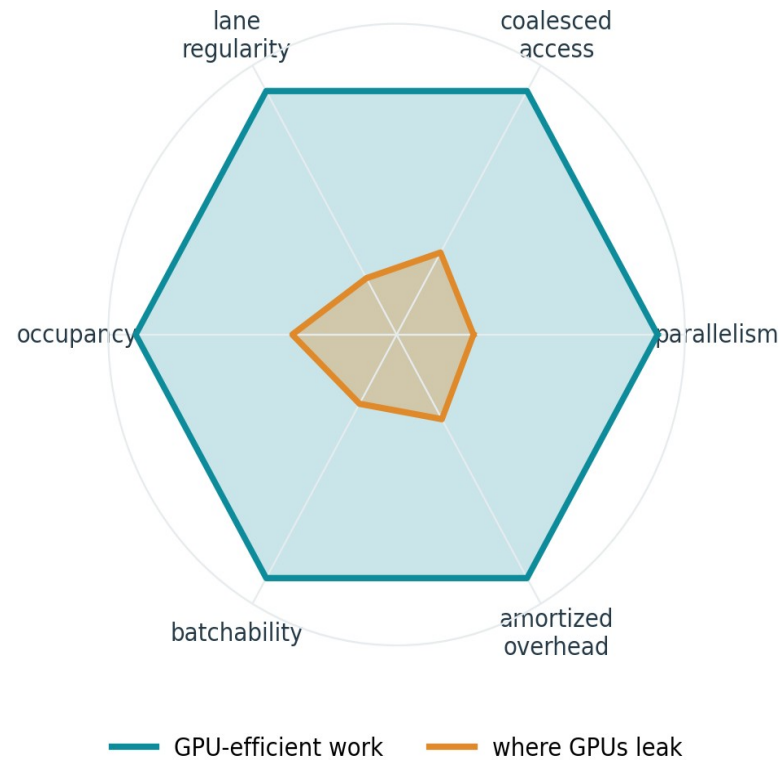
CPU and GPU spend silicon differently



- CPU: latency across diverse code.
- GPU: throughput across regular work.
- Neither wins every phase.

⇒ Specialization narrows the job in exchange for more useful work per area and per joule.

The GPU efficiency envelope is real and bounded



⇒ GPUs are unbeatable inside the envelope and leak outside it — that boundary is where specialists live.

02

The workload pressures

The measured bounds that reshape silicon

Signatures, not labels

Workload signatures, not labels

Training matmul	I	I	I	I	I	-
Transformer decode	-	lo	lo	I	lo	lo
Embedding lookup	-	-	I	-	hi	-
Graph aggregation	-	-	hi	-	lo	lo
Robotic planning	lo	lo	lo	lo	-	I
	arith. intensity	locality	parallelism	regularity	batchable	deadline

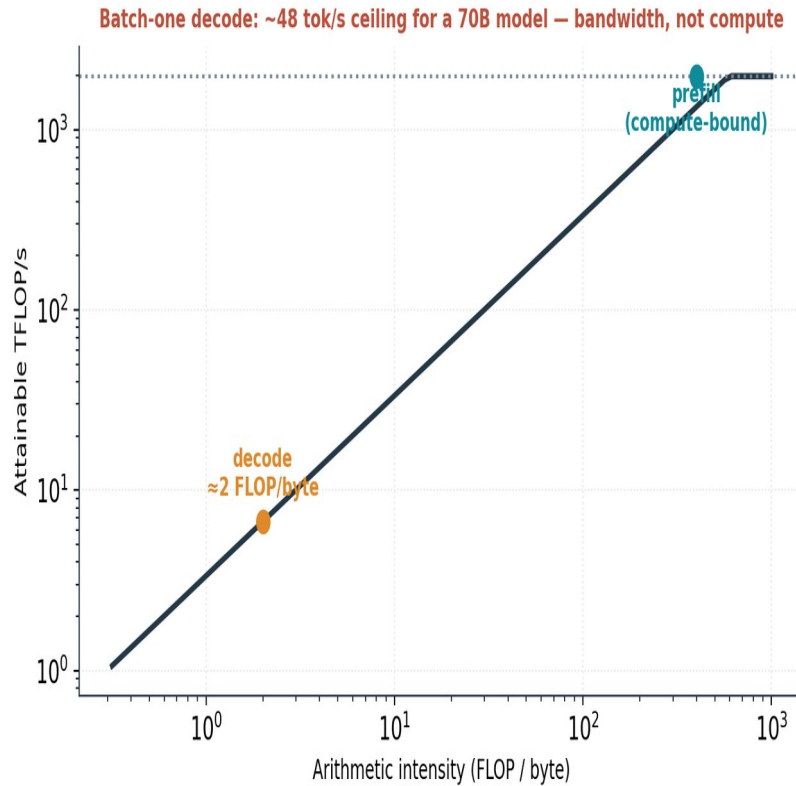
⇒ Characterize each phase by its behavior — the same model is compute-, bandwidth-, and capacity-bound in one request.

Prefill and decode want different machines



⇒ One pooled device compromises; phase-specialized silicon is now reaching hyperscale roadmaps.

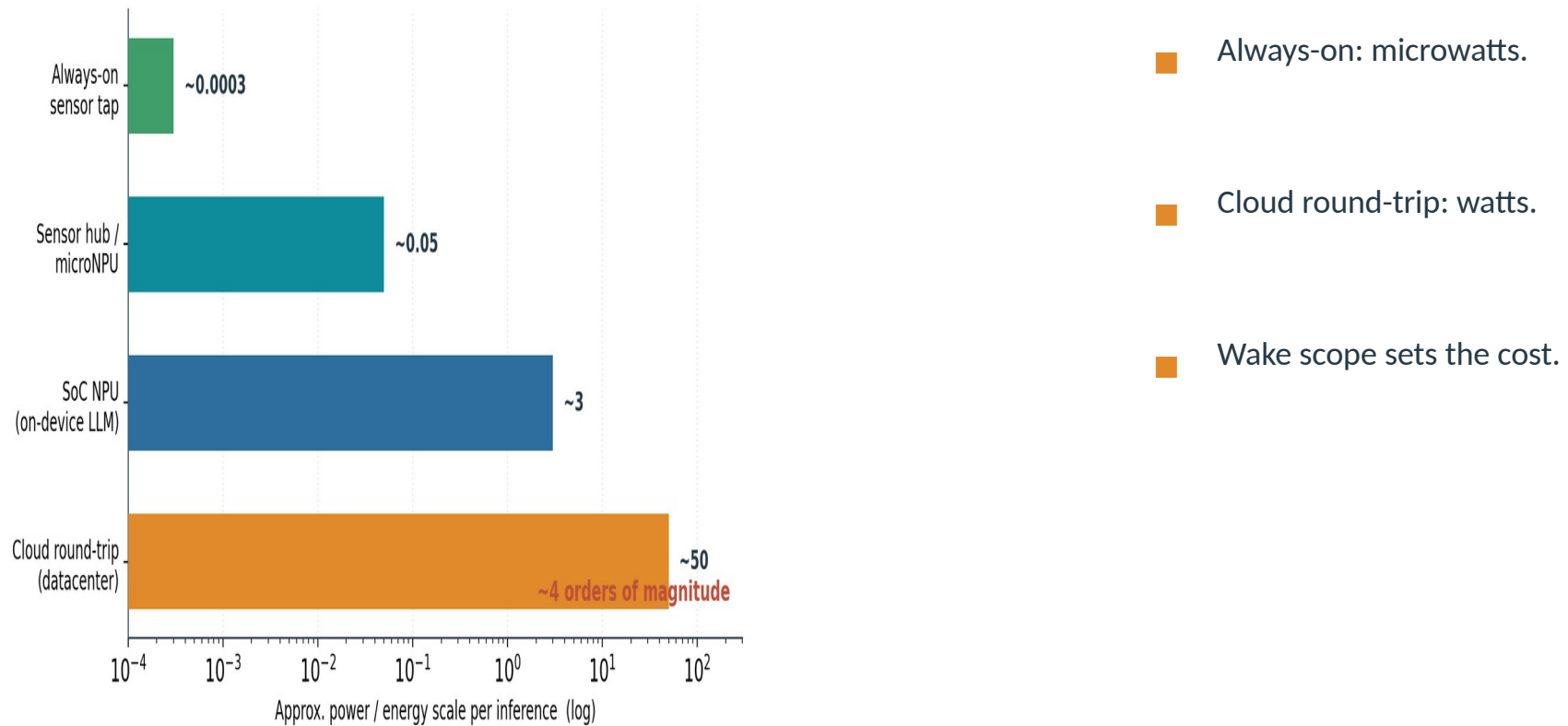
Batch-one decode is a bandwidth wall



- 70B model, 8-bit, one sequence.
- ≈48 tokens/s ceiling on an H100.
- ~0.3% of matrix throughput used.

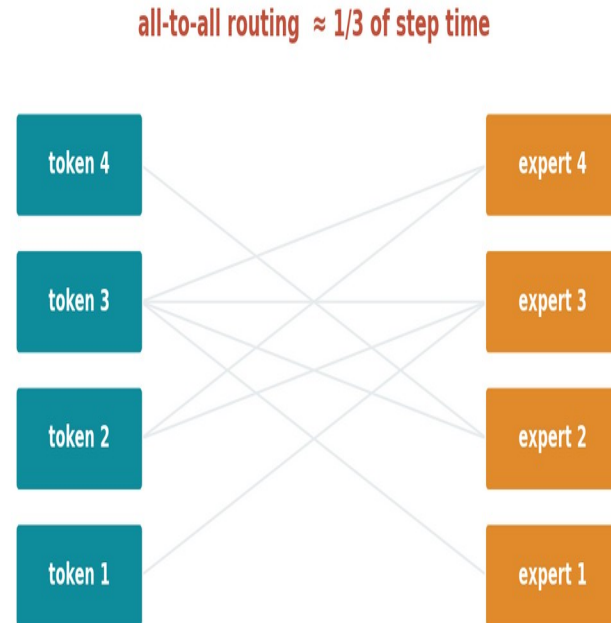
⇒ Only batching or a changed memory contract moves this — the pressure behind inference-first silicon.

Energy spans four orders of magnitude



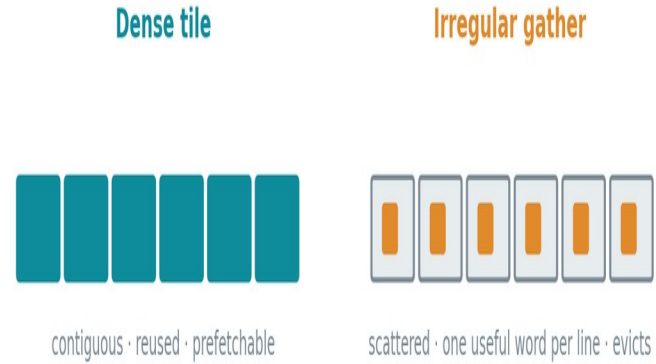
⇒ Always-on features are impossible on a GPU contract — a structural bound, not an efficiency gap.

MoE turns compute into communication



⇒ All-to-all routing can be $\sim 1/3$ of a training step — a bound in the fabric, not the arithmetic.

Irregular access defeats the cache



⇒ Embeddings, gather, and graph traversal are the bound no family yet owns cleanly.

Software succeeds — until it meets physics

3×

IO-aware attention throughput
(FlashAttention)

2–4×

KV paging / continuous batching
(vLLM)

~34%

of a training step in MoE all-to-all

<300 μ W

always-on keyword spotting
budget

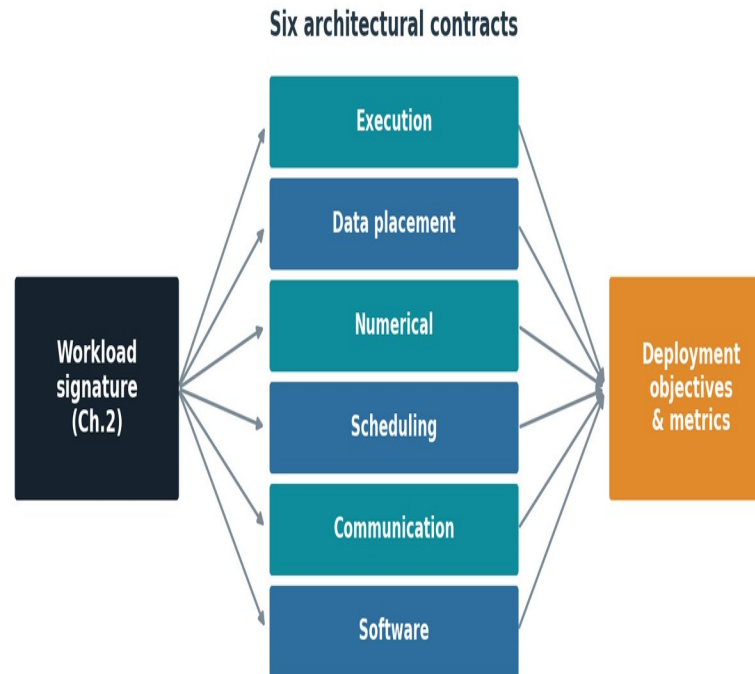
⇒ Each gain is real and each exposes a residual bound software cannot move — that residue is the case for hardware.

03

The decision framework

Contracts, and a test for when to specialize

Hardware defines six contracts



⇒ A signature stresses some contracts, not others — that mismatch is where a new architecture earns its place.

Four conditions — all must hold

1

Residual bound

A limit remains after software has done its best.

2

Material consequence

The bound actually hurts under the deployment's objective.

3

Effective lever

Hardware changes the bound rather than relocating it.

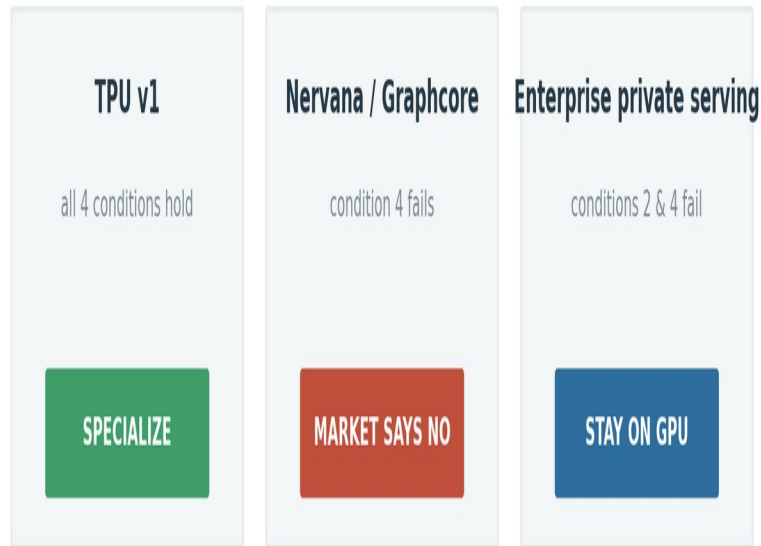
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Lifetime economics

Recurring benefit repays fixed cost, software, and lost flexibility.

⇒ Conditions 1-3 are the architect's analysis; condition 4 is the business case — and it decides most outcomes.

Same architecture, opposite verdicts



⇒ TPU v1 specialized; Nervana/Graphcore were right on architecture but failed condition 4; enterprise stays on GPU.

What the GPU absorbs — and what it cannot

Figure 5.5 — Partition of residual bounds by absorbability
Residual bounds from Chapter 2, partitioned by whether GPU evolution can absorb them without abandoning the platform contract (host-driven submission, SIMT programmability, shared software capital)

ABSORBED INTO THE GPU PLATFORM (evidence class A)	CONTESTED MIDDLE (decided by §2.14 condition 4)	STRUCTURALLY OUTSIDE THE THROUGHPUT CONTRACT
Matrix density → tensor cores	Low-batch decode bandwidth bound	Wake-scope / always-on energy (μW vs W domains)
Precision → FP16/BF16/FP8 engines	MoE routing / all-to-all (~1/3 step time)	Provable worst-case timing and certification
Structured sparsity → 2:4 paths	Near-memory embedding lookup	Near-sensor locality
Movement overhead → async copy / TMA	Phase-disaggregated serving	Embedded per-unit cost floors
Launch overhead → graph capture	GPU trajectory + software capital vs. contract-changing designs	Secure basis of the beyond-GPU case
Scale-up comms → NVLink domains		
Isolation → device partitioning		

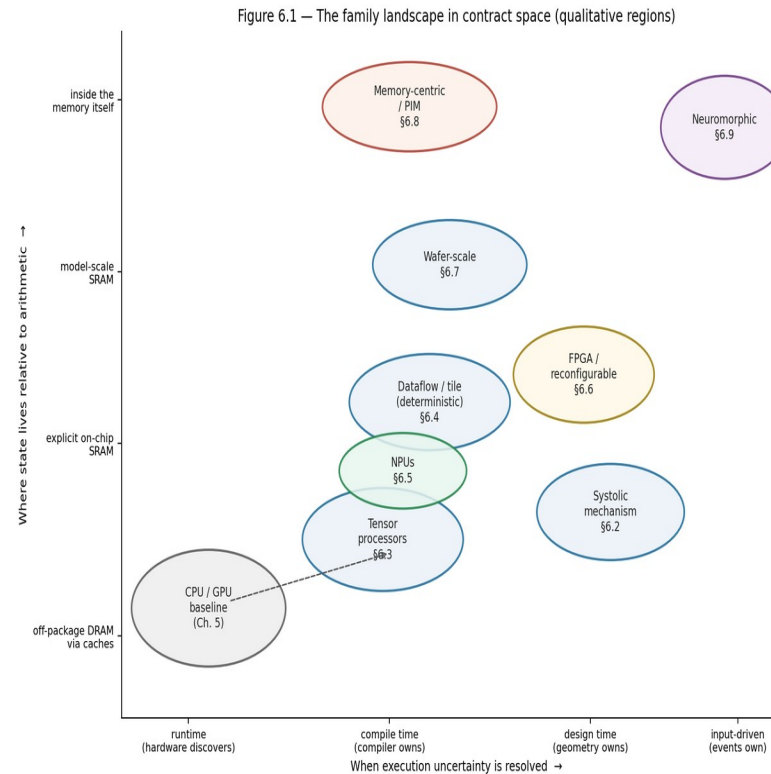
⇒ Absorbable bounds join the platform; structural bounds (wake scope, determinism, near-sensor, cost floors) do not.

04

The accelerator families

Eight contract changes, two organizing axes

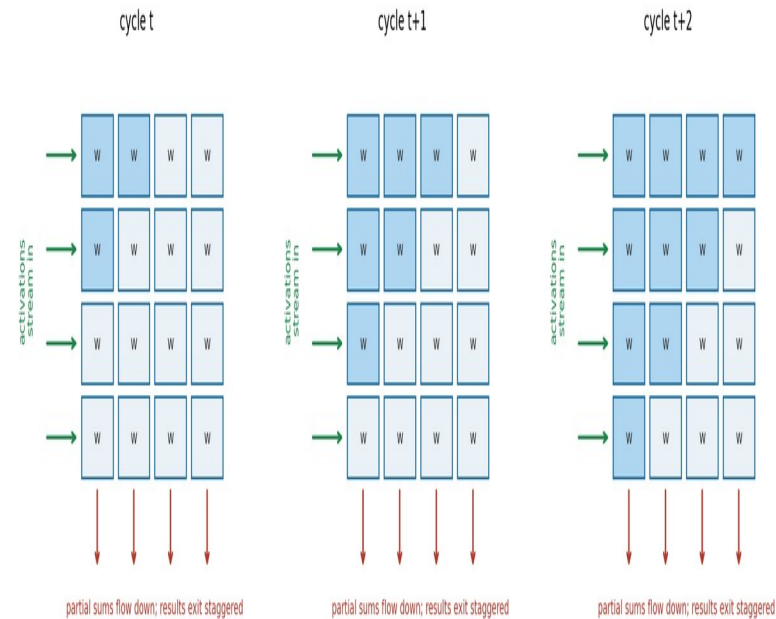
When uncertainty resolves × where state lives



⇒ Every family is a displacement from the CPU/GPU baseline on these two axes — no point dominates.

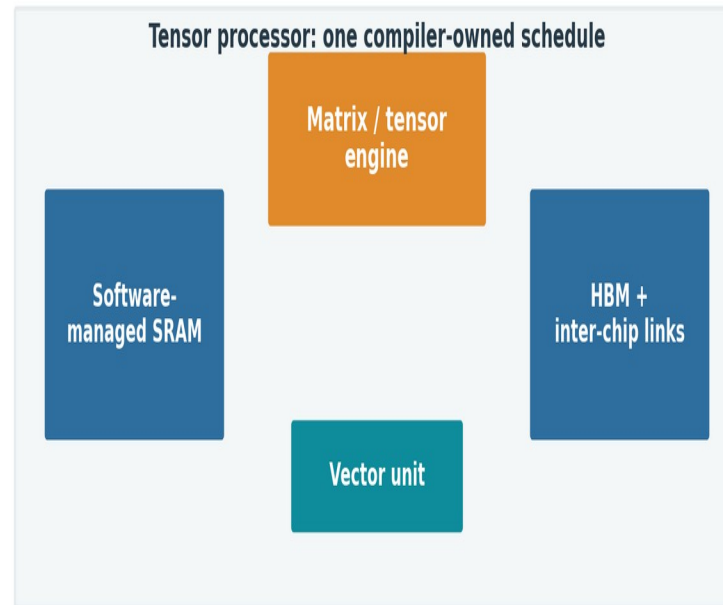
Systolic arrays: reuse from geometry

Figure 6.2 — Weight-stationary systolic execution: weights resident (w), activation wavefront advances one hop per cycle; every fetched value is reused across the array



⇒ A value fetched once is reused across the array — memory traffic only at the edges. Absorbed by everyone, including the GPU.

Tensor processors: one compiler-owned schedule



⇒ **Matrix engine + software-managed SRAM. Wins dense portfolios at owner scale (TPU, Trainium, MTIA).**

Deterministic dataflow / tile processors

GPU: uncertainty discovered at run time

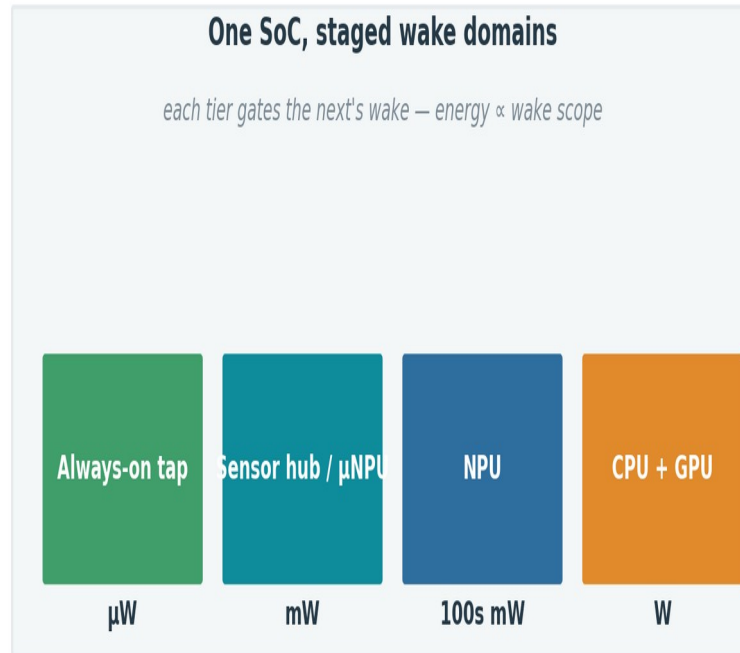


Dataflow / tile: schedule fixed at compile time



⇒ Schedule fixed at compile time → zero variance. Bids for latency-bounded decode and reproducible execution (Groq).

NPU and near-sensor engines



- Power-domain-first design.
- Value = wake-scope energy.
- Coverage, not TOPS, decides.

⇒ The most secure beyond-GPU territory: wake scope and cost floors are structurally outside the GPU contract.

FPGAs: flexibility after manufacturing

Reconfigurable fabric: two compilation speeds

Hardware compile

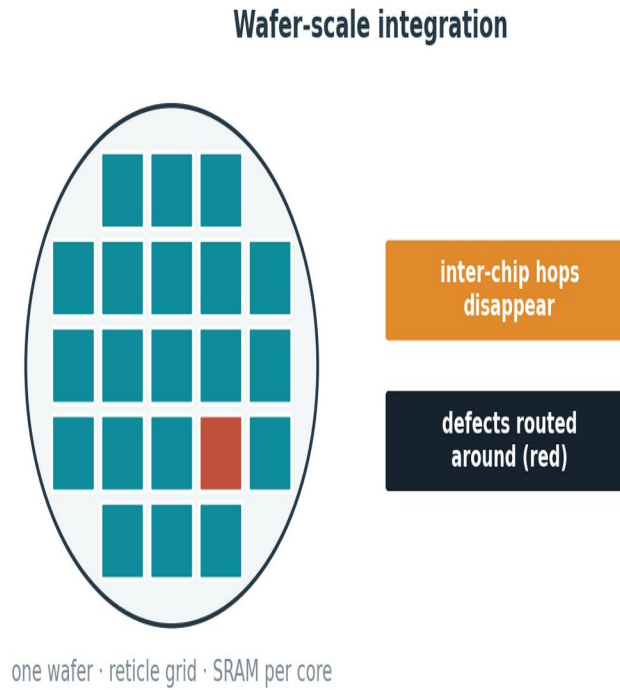
hours · max efficiency
hardware skills

Overlay / soft NPU

seconds · lower efficiency
software skills

⇒ Two compilation speeds. Wins where post-manufacture structure is the requirement; elsewhere a staging tier to ASIC.

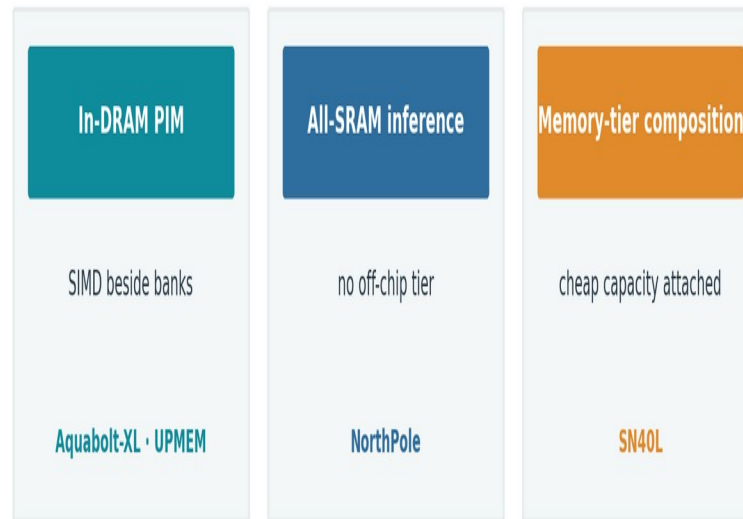
Wafer-scale: state at SRAM distance



⇒ Inter-chip communication becomes on-wafer routing; model-scale state stays close. Survives on anchor tenants.

Memory-centric / processing-in-memory

Compute moves to the data – three flavors



⇒ Strongest condition-3 case in the survey — gated only by the lack of a portable programming model.

Neuromorphic: energy proportional to activity



Mechanism

Event-driven spikes; synaptic state beside the neurons; nothing switches when nothing changes.



Evidence

Research-class (TrueNorth, Loihi). Validates the energy model, not yet a mainstream workload.

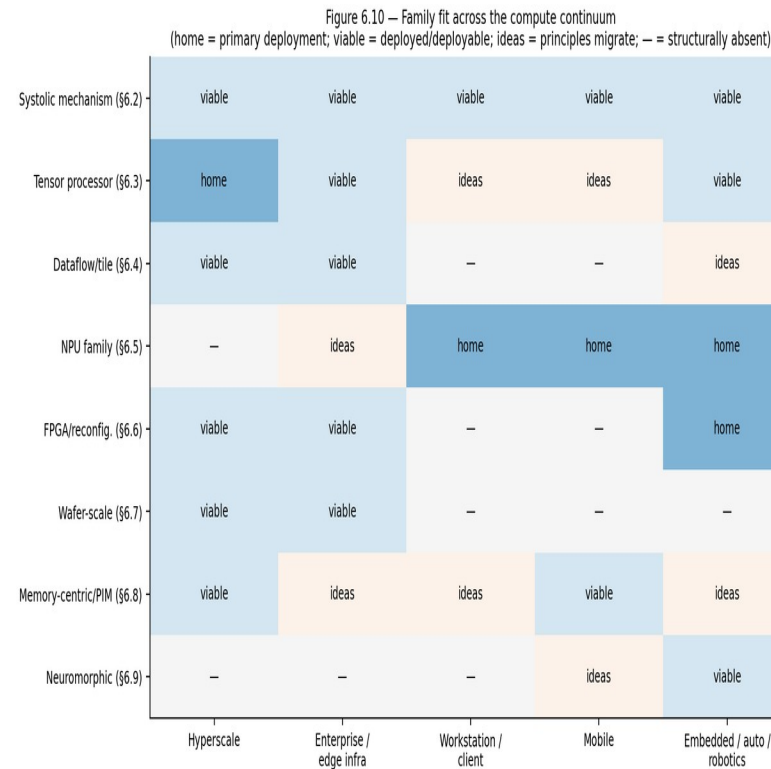


Beachhead

Extreme edge: always-on sensory, reflex loops, event cameras — sparse-in-time signals.

⇒ The theoretical endpoint of wake-scope logic — a direction of development, not yet a destination.

Same families, different binding constraints



⇒ Ideas migrate both ways; implementations diverge because each deployment prices the stressors differently.

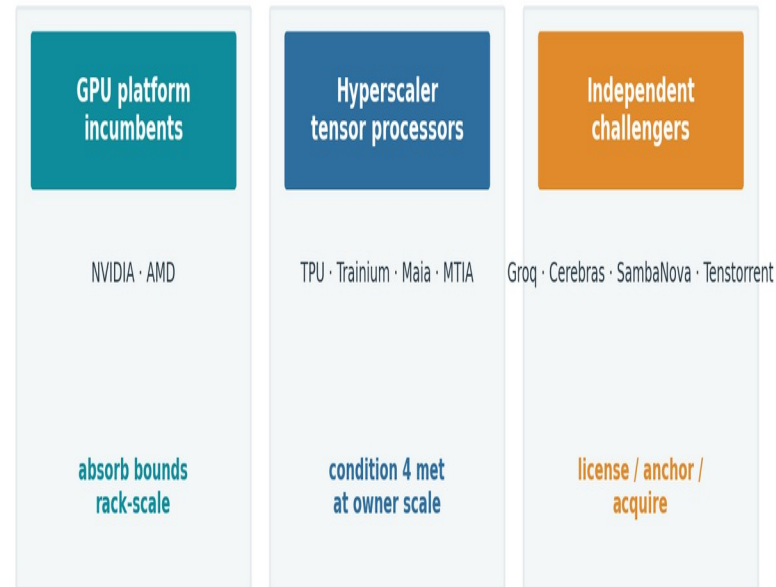
05

The landscape, grouped

Read through the framework — not a product catalog

Three groups, one deciding variable

Grouped by how condition 4 resolves — not a product catalog



⇒ None of the challengers failed on architecture — every trajectory was decided by condition 4.

The TPU verdict, replicated four times



Google TPU

Seventh generation; inference-emphasized; pods of thousands.



AWS Trainium

Generally available; proprietary switch fabric; 'token economics.'



Microsoft Maia

Inference-first; large on-chip SRAM; standard Ethernet.

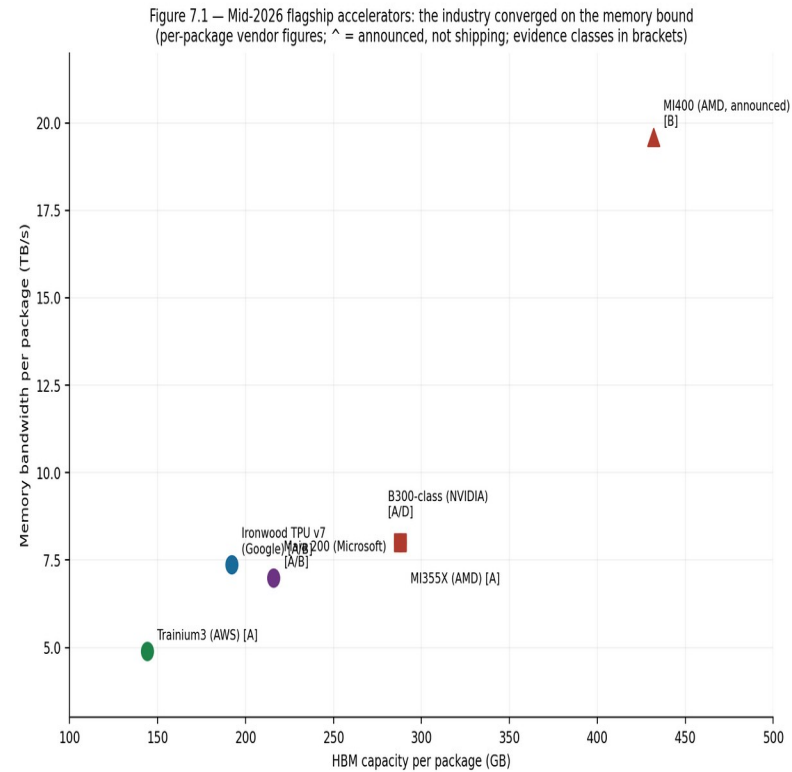


Meta MTIA

Recommendation-first; peer-reviewed first generation.

⇒ Every major cloud fields internal tensor processors at generation depth — the second transition as industrial fact.

The competition is a memory race



⇒ Flagships cluster on capacity and bandwidth, not FLOPS — the absorption thesis operating in public.

The contested middle consolidates



License in

A deterministic-dataflow architecture licensed into the incumbent platform.



Anchor tenancy

Wafer-scale survives on a few enormous compute agreements, then IPO.



Acquisition

Programmable tile processors absorbed by a larger incumbent.

⇒ **Condition 4 — software capital, volume, distribution — decided each, not architectural merit.**

The continuum's other half



AI PC

NPU + GPU under shared memory; bandwidth and coverage, not TOPS, decide.



Mobile SoC

Three wake tiers; on-device foundation models; thermal equilibrium separates the field.



Embedded / μ NPU

IP-licensed; cost floor and decade lifecycle dominate.



Robotics

GPU-class perception above a deterministic control floor — flexibility wins when algorithms churn.



Automotive

Determinism and lifecycle become certification objects; supply sovereignty joins economics.



Shared taxonomy

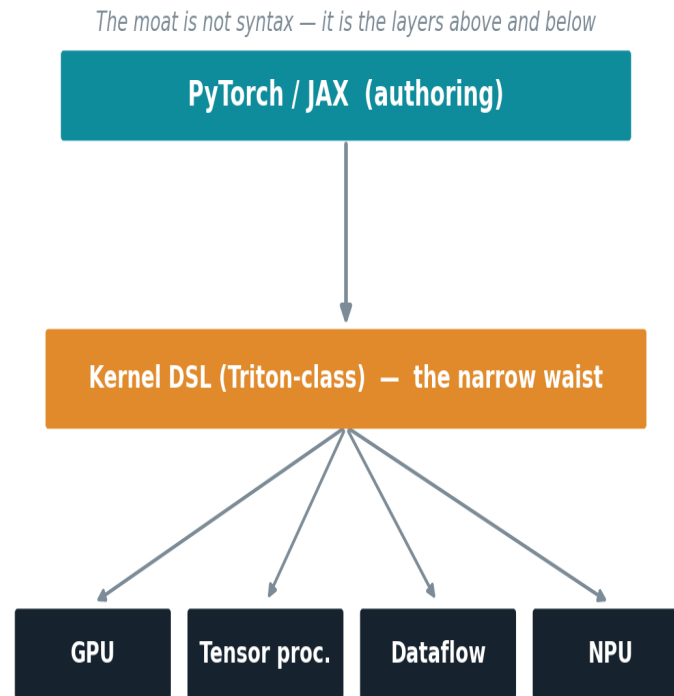
Same families as the datacenter — different binding constraints.

06

Substrate & economics

Software capital, packaging, and cost

The moat is not the kernel language



⇒ Portability efforts aimed at the kernel layer solve the cheapest part; the moat sits above and below the waist.

Integration is the frontier



HBM

Sets every bandwidth bound; three-vendor supply; rivals the logic die in cost share.



Packaging

Reticle limit drives 2.5D, chiplets, hybrid bonding; capacity itself gates supply.

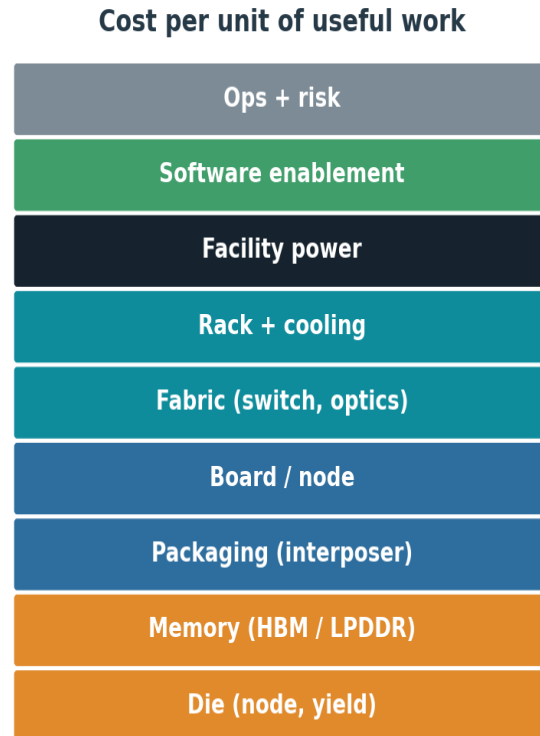


Fabric & optics

Scale-up vs Ethernet; co-packaged optics attacks network power, not latency.

⇒ The competitive frontier moved from logic to integration — the package, the rack, the facility.

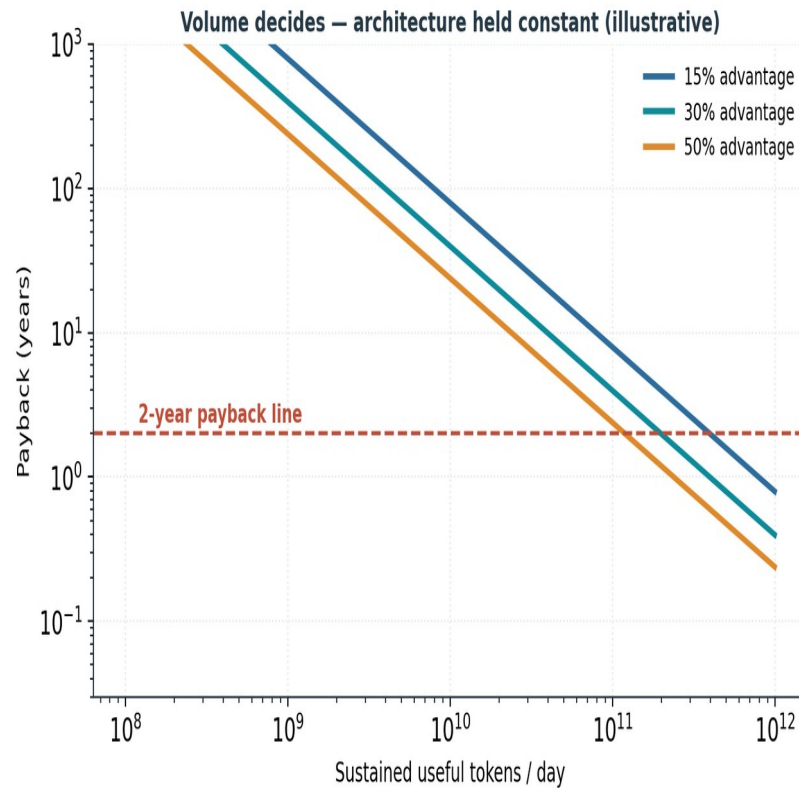
The full cost stack of useful work



- Every family redistributes weight.
- None shrinks the stack uniformly.
- Power now gates deployments.

⇒ The necessity test, in cost terms: does a redistribution lower the total for this workload at this volume?

Volume decides — architecture held constant



- Fixed cost $\approx$ software enablement.
- Savings scale with volume.
- Flexibility has option value.

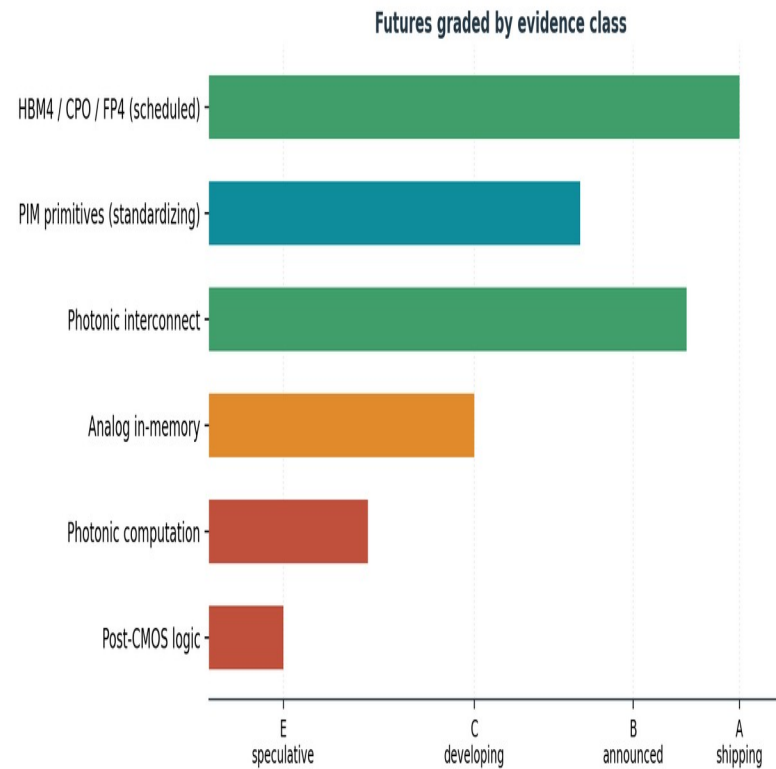
⇒ Below the volume threshold no efficiency percentage rescues the case; above it, small percentages justify large programs.

07

Futures & conclusions

Graded by evidence; what holds everywhere

Graded by evidence class



⇒ New memories arrive before new logic; the nearest contract changes enter inside incumbent platforms, via packaging.

What could move the bounds first



Test-time compute

Reasoning longer, not larger — deepens the decode bound, strengthens inference silicon.



Persistent-state agents

Promote state residency and freshness to first-class bounds — a possible new family.



Default sparsity

MoE and retrieval push routing and gather up the priority list.

⇒ If the signatures move, the family taxonomy should be re-derived — the method predicts its own revision.

What holds across the whole picture

A

Software converges on physics

Both transitions are software succeeding until it exposes a hard, measurable bound.

B

Necessity is decidable

A four-condition test returns specialize / market-no / stay-on-GPU from measurable inputs.

C

Condition 4 dominates

Software capital, volume, and distribution decide outcomes more than silicon merit.

D

Families, not a successor

Each changes a contract the GPU cannot adopt without ceasing to be a GPU.

E

Bounds moved to memory & power

Capacity, bandwidth, fabric, and facility power separate contenders — not FLOPS.

F

One continuum

The same taxonomy reads microwatt silicon and megawatt pods.

How to use the framework



Software architects

Design against signatures; respect batch-one; measure at the useful-result boundary; the test, not the catalog.



Platform architects

Evaluate at pod and facility; prefer consumption to ownership until volume proves otherwise; protect software seams.



Hardware selection

Signatures → software ceiling → locate the bound in the partition → weight the stack → run the test.

THE ARGUMENT IN ONE SENTENCE

Software optimization succeeds until it exposes a bound; the bound is absorbed by the GPU platform, escaped by changing a contract, or lived with — and the deployment class decides which.

Everything else — signatures, contracts, families, economics — is the machinery that makes that choice decidable.